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Censored Regression: Local Linear Approximations and Their Applications

Jianqing FAN and Irène GIJBELS*

Various statistical tools are available for modeling the relationship between response and covariate if the data are fully observable. In the situation of censored data, however, those tools are no longer directly applicable. This article provides an easily implemented methodology for modeling the association, based on censored data. The form of the regression relationship will be completely determined by the data; no assumptions are made about this form. Basic ideas behind the methodology are to transform the observed data in an appropriate simple way and then to apply a locally weighted least squares regression. The proposed estimator involves a variable bandwidth that automatically adapts to the design of the data points. That the methodology is very easy to implement is illustrated by several examples, including simulation studies and an analysis of the Stanford Heart Transplant Data and the Primary Biliary Cirrhosis Data. Several theoretical considerations are reflected in the examples. Finally, some basic asymptotic results are established.

KEY WORDS: Adaptive variable bandwidth; Asymptotic normality; Censoring; Cross-validation; Transformation.

1. INTRODUCTION

Regression techniques are commonly used for exploring the relationship between response and covariate. In the non-censored case, there are quite a few statistical tools for modeling this relationship. For example, the scatterplot can suggest whether the model is linear, quadratic, or of any other specific parametric form. But for censored data, these techniques are not directly applicable. Indeed, some of the responses and/or covariates may not be observed—they may be censored. This article's main purpose is to discuss an easily implemented and robust method for modeling the association between the response and the covariate when the response is censored. Although we devote our attention to the univariate case, we would like to mention that the basic ideas of our methodology hold for the case of two or more covariates.

Buckley and James (1979) introduced an appealing idea for the censored regression when the regression function is linear. A first attempt to investigate the consistency of the Buckley–James estimator was made by James and Smith (1984). A slight modification of the estimator led Ritov (1990) and Lai and Ying (1991) to establish the asymptotic normality of this type of estimators. Issues of efficiency and robustness were discussed by Ritov (1990). A family of asymptotically equivalent estimators, based on linear rank tests for the slope in the linear model, was introduced by Tsiatis (1990).

It is usually not known whether or not the model is linear. Scatterplots and residual plots are important diagnostic tools for uncensored regression. But in the censored case, these tools are not directly applicable. One may transform the data in an unbiased way to account for the censoring. An example

of this kind was given by Buckley and James (1979); however, their transformation involves the unknown regression function leading to an iterative scheme. Motivated by the Buckley–James transformation, Koul, Susarla, and Van Ryzin (1981) considered a transformation that depends only on the censoring distribution (assumed to be independent of the covariate), not on the regression function. Zheng (1987) proposed a class of transformations of this type. Once such a transformation is carried through, one can apply a variety of statistical techniques to analyze the transformed data as if they were uncensored. But because such a transformation does not involve the distribution of the response variable, it increases the variability. Hence some smoothing technique is necessary for modeling the transformed data. Some nonparametric regression techniques were applied by Doksum and Yandell (1982), Dabrowska (1987), and Zheng (1988).

There is a vast literature in nonparametric regression (Eubank 1988; Härdle 1990; Müller 1988; Wahba 1990). This article focuses on local linear regression smoothers, which are simple and easily understood. Among various competitive regression estimators proposed in the literature, Fan (1992) showed that the local linear regression smoother repairs the drawbacks of other types of kernel estimators, such as Nadaraya–Watson (Nadaraya 1964; Watson, 1964) and Gasser–Müller (Gasser and Müller 1979) estimators. This method adapts to various types of designs—random and fixed, highly clustered and nearly uniform—and reduces boundary effects. Decision-theoretical considerations emphasize the superiority of the local linear approximation method (see Fan 1993). We use an adaptive (variable) bandwidth, which is large or small depending on the sparsity of the design points. This bandwidth, coupled with the local approximation idea, yields a viable method for the censored regression situation.

The article is organized as follows. Section 2 summarizes the proposed methodology. Section 3 shows how the observed data can be transformed in an appropriate way. Section 4

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presents the local linear regression smoother. Section 5 provides some basic asymptotic results of the proposed procedure, with proofs given in the Appendix. Finally, Section 6 illustrates the performance of the proposed procedure via simulation studies and via analysis of data sets such as the Stanford Heart Transplant data and the Primary Biliary Cirrhosis data.

2. SUMMARY OF THE METHODOLOGY

Consider the regression model $Y = m(X) + \sigma(X)\varepsilon$, where Y is the survival time, X is the associated covariate, the function $m(\cdot)$ is the unknown regression curve, and $\sigma(\cdot)$ is the conditional variance representing the possible heteroscedasticity. Assume that X and ε are independent, $E\varepsilon = 0$ and $\text{var}(\varepsilon) = 1$. Let Y and C be conditionally independent random variables given X , where C is the censoring time associated with the survival time Y . Denote $Z = \min(Y, C)$ and $\delta = I(Y \leq C)$. The observations are $\{(X_i, Z_i, \delta_i): i = 1, \dots, n\}$, which is a random sample from the population (X, Z, δ) . Throughout this article, the observations are ordered according to the X_i 's.

The method for estimating $m(\cdot)$ consists of the following steps. In the subsequent sections we discuss and explain each step in detail.

Transformation of the data. For a given integer k and a nonnegative weight function K , replace the observed data $\{(X_i, Z_i, \delta_i)\}$ by $\{(X_i, Y_i^*)\}$, where

$$Y_i^* = \delta_i Z_i + (1 - \delta_i) \frac{\sum_{j: Z_j > Z_i} Z_j K\left(\frac{X_i - X_j}{(X_{i+k} - X_{i-k})/2}\right) \delta_j}{\sum_{j: Z_j > Z_i} K\left(\frac{X_i - X_j}{(X_{i+k} - X_{i-k})/2}\right) \delta_j}. \quad (1)$$

Application of the local linear regression technique. Starting from the transformed data $\{(X_i, Y_i^*)\}$, do a local linear regression using the same smoothing parameter k . Namely, compute

$$\hat{m}(x) = \frac{\sum_{i=1}^n w_i(x) Y_i^*}{\sum_{i=1}^n w_i(x)}, \quad (2)$$

with

$$w_i(x) \equiv K\left(\frac{x - X_i}{\hat{h}_k(x)}\right) [s_{n,2} - (x - X_i)s_{n,1}],$$

where

$$s_{n,l} = \sum_{i=1}^n K\left(\frac{x - X_i}{\hat{h}_k(x)}\right) (x - X_i)^l, \quad l = 0, 1, 2.$$

Here we use the adaptive variable bandwidth of order k ,

$$\hat{h}_k(x) = (X_{l+k} - X_{l-k})/2, \quad (3)$$

where l is the index of the design point X_l closest to x .

Selection of the smoothing parameter k by cross-validation. For a given k , transform the data by (1), then disregard the i th transformed data point (X_i, Y_i^*) and denote by $\hat{m}_{-i}(x)$ the estimator calculated from (2). Compute $CV(k) = \sum_{i=1}^n (Y_i^* - \hat{m}_{-i}(X_i))^2$. Denote by $\hat{k}$ the cross-validation

smoothing parameter; that is, the parameter that minimizes $\{CV(k): k = 1, \dots, [(n-1)/2]\}$, where $[a]$ is the greatest integer part of a .

Note that another possible approach is to delete the i th observation before carrying through the transformation step. But this would result in an extra loop operation, which would increase the computation cost by approximately a factor n . Hence this approach is not recommended.

Calculation of the local linear smoother. With $\hat{k}$ as a smoothing parameter, transform the observed data by (1) and use the smoother (2) to estimate the regression function.

2.1 Remarks Concerning Computations and Implementation

The aforementioned cross-validation step requires very intensive computations. An alternative is choosing the bandwidth by "trial and error." But in both cases it is desirable to have a fast way of computing the estimated regression curve, such as linear binning (see Silverman 1986). Consider a generic variable bandwidth $h(x)$, possibly depending on the data. Suppose that we are interested in estimating the regression function $m(\cdot)$ on an interval $[a, b]$ at grid points $\{x_j = a + j\Delta, j = 0, \dots, g\}$, with $\Delta = (b - a)/g$. The basic idea is to conceptually move each observation (X_i, Y_i^*) toward its two nearest grid points and to assign weights to each of the two new points according to the corresponding distance between the original point and the grid point. Formally, if $X_i \in [x_j, x_{j+1})$, move the point (X_i, Y_i^*) to the points (x_j, Y_i^*) and (x_{j+1}, Y_i^*) and associate weights $(x_{j+1} - X_i)/\Delta$ and $(X_i - x_j)/\Delta$. Now apply the local linear regression method (see Sec. 4) to the new data set. This leads to the "binned" version of the regression estimator given by

$$\hat{m}_b(x_p) = \frac{\sum_{j=0}^g Y_{j,b}^* K\left(\frac{(p-j)\Delta}{h(x_p)}\right) [s_{2,p} - (p-j)s_{1,p}]}{s_{0,p}s_{2,p} - s_{1,p}^2},$$

for $p = 0, \dots, g$. Here, with

$$W_{i,j} = (1 - |X_i - x_j|/\Delta)_+, \quad Y_{j,b}^* = \sum_{i=1}^n W_{i,j} Y_i^*$$

and

$$s_{q,p} = \sum_{j=0}^g K\left(\frac{(p-j)\Delta}{h(x_p)}\right) (p-j)^q \left(\sum_{i=1}^n W_{i,j}\right), \quad q = 0, 1, 2.$$

We remark that with a variable bandwidth, as for example in (3), the foregoing regression smoother does not admit a convolution form. But Fan and Marron (1994) reported that the time saving from the "binned implementation" (compared to the direct approach) does not come from the convolution form, for a good resolution $g = 400$, but rather from its simpler structure. Thus, the binning idea can also reduce the computational cost in the current setting.

3. TRANSFORMATION OF THE DATA

Denote by $\phi_1(\cdot, \cdot)$ and $\phi_2(\cdot, \cdot)$ the transformation functions acting on the uncensored and censored observations.

More precisely, replace the data point (X, Z, δ) with the transformed data point (X, Y^*) according to

$$\begin{aligned} Y^* &= \phi_1(X, Y) \quad \text{if uncensored} \\ &= \phi_2(X, C) \quad \text{if censored} \\ &= \delta\phi_1(X, Z) + (1 - \delta)\phi_2(X, Z). \end{aligned} \quad (4)$$

In the sequel of this article we will refer to this transformation as the "ideal transformation," because it assumes that the transformation functions $\phi_1(\cdot, \cdot)$ and $\phi_2(\cdot, \cdot)$ are known. In practical situations, however, those transformation functions typically must be estimated. At that stage the dependence on the data points will be showing up in the estimated transformations; see, for example, (7).

Because the inference will be based on (X, Y^*) , the basic requirement is that $E(Y^*|X) = m(X)$, ensuring that we estimate the right object. This requirement is equivalent to

$$\begin{aligned} E\{(\phi_1(X, Y)\bar{G}(Y|X) + \phi_2(X, C)\bar{F}(C|X))|X\} \\ = E(Y|X), \end{aligned}$$

where $\bar{F}(\cdot|x)$ and $\bar{G}(\cdot|x)$ are the conditional survival function of the random variables Y and C given $X = x$. To avoid technicalities, we assume that $\bar{G}(\cdot|x)$ is continuous, and for simplicity we work with nonnegative random variables. Then the preceding equation can be written as

$$\begin{aligned} \int_0^{+\infty} \left[\phi_1(x, y)\bar{G}(y|x) - \int_0^y \phi_2(x, c) d\bar{G}(c|x) - y \right] \\ \times d\bar{F}(y|x) = 0 \end{aligned} \quad (5)$$

for all x and y . The quality of the transformation is reflected in $\text{var}(Y^*|X = x)$.

An intuitive and appealing transformation was given by Buckley and James (1979):

$$\phi_1(x, y) = y \quad \text{and} \quad \phi_2(x, y) = E(Y|Y > y, X = x). \quad (6)$$

This transformation can be written as $Y_0^* = E(Y|\delta, Z, X)$. It is closest to the original response in the sense that $E(Y - Y_0^*)^2 \leq E(Y - Y^*)^2$; thus it can be regarded as the "best restoration." But this transformation depends on the unknown regression function leading to an iterative procedure. The computation for such a procedure would be formidable, especially when using nonparametric techniques with a cross-validated smoothing parameter. Besides, computing the conditional expectation in (6) requires strong model assumptions, limiting the applicability. A fast and dynamical transformation requires estimates with an explicit form, such as a local average. We now offer two possible approaches.

The first approach, inspired by (6), is to transform the data as follows. Suppose that the point (X_i, Z_i, δ_i) is a censored one (i.e., $\delta_i = 0$). Consider a neighborhood of X_i , and replace the censored observation Z_i by the weighted average of all uncensored responses in that neighborhood that are larger than Z_i . Formally, let k be an integer that will determine the neighborhood of X_i , and denote by K a nonnegative weight function. We recall that all observations $\{(X_i, Z_i, \delta_i): i = 1, \dots, n\}$ are ordered according to the X_i 's. Define

$$Y_i^* = \frac{\sum_{j: Z_j > Z_i} Z_j K\left(\frac{X_i - X_j}{(X_{i+k} - X_{i-k})/2}\right) \delta_j}{\sum_{j: Z_j > Z_i} K\left(\frac{X_i - X_j}{(X_{i+k} - X_{i-k})/2}\right) \delta_j}, \quad (7)$$

with the usual convention of truncating the neighborhood when it is out of the data range. This transformation is a nonparametric estimate of the conditional expectation in (6). Uncensored data points remain unchanged (i.e., $Y_i^* = Z_i$, if $\delta_i = 1$). The transformed data are $\{(X_i, Y_i^*): i = 1, \dots, n\}$. Because transformation (6) is the "best restoration," the estimated transformation (7) is favorable and intuitively appealing.

A second approach is to choose transformations that do not explicitly depend on $m(\cdot)$, according to the following procedure. Consideration of (5) leads to the requirement that

$$\begin{aligned} \phi_1(x, y)\bar{G}(y|x) - \int_0^y \phi_2(x, c) d\bar{G}(c|x) - y = 0, \\ \forall x \text{ and } y. \end{aligned} \quad (8)$$

This clearly results in transformations that depend only on the conditional censoring distribution function. Thus these transformations are relatively easy to implement, especially when the conditional censoring distribution is independent of the covariate. But a price—in terms of increased variance—must be paid, as will be seen from the simulated example. In the sequel we discuss some simple transformations obtained via this approach.

Put $\phi_2(x, y) = \phi_1(x, y) + d(x, y)$. Then the solution to (8) can be expressed as

$$\begin{aligned} \phi_1(x, y) &= \int_0^y \bar{G}^{-1}(t|x) dt \\ &\quad + \int_0^y d(x, t)\bar{G}^{-1}(t|x) d\bar{G}(t|x), \\ \phi_2(x, y) &= \int_0^y \bar{G}^{-1}(t|x) dt \\ &\quad + \int_0^y d(x, t)\bar{G}^{-1}(t|x) d\bar{G}(t|x) + d(x, y), \end{aligned} \quad (9)$$

which was also given by Zheng (1987) from a different motivation. Different choices of the function $d(\cdot, \cdot)$ result in a different degree of variability. Imposing $d(x, 0) = 0$ for all x , a straightforward but tedious calculation shows that

$$\begin{aligned} \text{var}(Y^*|X = x) &= \sigma^2(x) + \int_0^{+\infty} \left[2 \int_0^y \int_0^v \frac{G(s|x)}{\bar{G}(s|x)} ds dv \right. \\ &\quad \left. + \int_0^y \left\{ \frac{2(y-v)d(x, v)}{\bar{G}(v|x)} - d^2(x, v) \right\} d\bar{G}(v|x) \right] dF(y|x); \end{aligned}$$

this quantity measures the variability of the transformation. Clearly, the anti-intuitive choice $d < 0$ is inadmissible, because it is dominated by the choice $d \equiv 0$.

A specific subclass of the transformation (9) is that $d(x, y) = \alpha y / \bar{G}(y|x)$, leading to the class of transformation

$$\begin{aligned}\phi_1(x, y) &= (1 + \alpha) \int_0^y \frac{dt}{\bar{G}(t|x)} - \alpha \frac{y}{\bar{G}(y|x)}, \\ \phi_2(x, y) &= (1 + \alpha) \int_0^y \frac{dt}{\bar{G}(t|x)}.\end{aligned}\quad (10)$$

Note that the Koul, Susarla, and Van Ryzin (1981) transformation (hereinafter referred to as the KSV transformation) corresponds to $\alpha = -1$ and that the Leurgans (1987) transformation relates to $\alpha = 0$. The tuning parameter α in this new class (abbreviated as NC) of transformations (10) creates the opportunity to improvement. The choice $\alpha > 0$ focuses more on the censored observations than on the uncensored observations, which is more intuitive than Leurgans's equal choice. On the other hand, when α is large, $\phi_1(x, y) < y$, causing more variability. This leads us to take the largest α such that $\phi_1(X_i, Y_i) \geq Y_i$ for the uncensored response:

$$\hat{\alpha} = \min_{\{i: \delta_i=1\}} \frac{\int_0^{Y_i} \frac{dt}{\bar{G}(t|X_i)} - Y_i}{\frac{Y_i}{\bar{G}(Y_i|X_i)} - \int_0^{Y_i} \frac{dt}{\bar{G}(t|X_i)}}. \quad (11)$$

This choice of α reduces the variability and will be used in our implementations.

In many applications, censoring is caused by the termination of study. Thus it is reasonable to assume that $G(c|x) \equiv G(c)$, which can be estimated by the Kaplan–Meier product-limit estimator. The advantage of transformation (10) is its simplicity—one reduces the censored regression to the usual regression even in the multivariate setting. When C and X are not independent, one can use Beran's (1981) local product-limit estimator for $\bar{G}(c|x)$, defined as

$$\hat{\bar{G}}_n(c|x) = \prod_{j: Z_j \leq c} (1 - \hat{\lambda}_j(x))^{1-\delta_j}, \quad (12)$$

where $\hat{\lambda}_j(x) = 1/\#\{i: Z_i \geq Z_j, i \in I_k(x)\}$, with $I_k(x)$ the set of indices of the $(2k+1)$ nearest neighbors of x , chosen in a symmetric way. The estimator for $\bar{G}(c|x)$ is inaccurate at the tail; to avoid this inaccuracy, we do not transform a particular data point when $\hat{\bar{G}}_n(Z_j|X_j) \leq c_k$. We will take $c_k = 1/k$ in our implementation.

4. LOCAL LINEAR REGRESSION SMOOTHERS

Starting from the transformed data $\{(X_i, Y_i^*): i = 1, \dots, n\}$, we now estimate the true regression function $m(\cdot)$. Recall that the data are ordered according to the X_i 's. Fix a point x and approximate the unknown function

$$m(z) \approx m(x) + m'(x)(z - x) \equiv a + b(z - x) \quad (13)$$

for z in a neighborhood of x . Thus, locally, the problem of finding $m(x)$ is equivalent to finding the intercept a of the approximating regression line. The local neighborhood here is determined by a kernel function K and a smoothing parameter h_n , which might depend on the location x and/or the data points. To avoid problems caused by the sparsity

of data points, it is preferable to use a smoothing parameter that adapts to the design of the data points.

The basic idea is to use a bandwidth (3). The parameter k determines the number of local data points. Intuitively, in a dense (sparse) design region, one should use a small (large) bandwidth (see also Thm. 5.1). The bandwidth (3) discretizes the constant bandwidth h in a natural way, with the flexibility of changing the degree of the design adaptation; that is, one can use, for example, $\hat{h}_k^{1/2}(x)$ or $\hat{h}_k^{1/4}(x)$. This discretization also makes it much easier to compute the optimal smoothing parameter. The optimal local bandwidth was discussed by Staniswalis (1989) in a more restricted setup. We do not use it here because it is too complicated in this generality.

With the local modeling (13), the problem of estimating $m(x)$ becomes minimizing

$$\sum_{i=1}^n (Y_i^* - a - b(X_i - x))^2 K\left(\frac{x - X_i}{\hat{h}_k(x)}\right)$$

with respect to a and b , resulting in the solution $\hat{a}$ and $\hat{b}$. The estimator for $m(x)$ based on the local linear approximation method is defined as $\hat{m}(x) = \hat{a}$. Note that $\hat{b}$ is an appropriate estimator for $m'(x)$. Straightforward algebra shows that

$$\hat{m}(x) = \sum_{i=1}^n w_i(x) Y_i^* / \sum_{i=1}^n w_i(x), \quad (14)$$

with the weights $w_i(x)$ given in (2). This type of estimation method was introduced by Stone (1977) and by Cleveland (1979). In the ordinary—noncensored—regression setup, this estimator possesses several nice properties, including best linear smoother in an asymptotic minimax sense, no requirement for boundary modifications, and adaptation to various types of designs (see Fan 1992, 1993 and Fan and Gijbels 1992).

5. ASYMPTOTIC RESULTS

We first provide insight into the behavior of the variable bandwidth $\hat{h}_k(x)$ defined in (3). The following theorem shows that this variable bandwidth behaves approximately as $k/(nf_X(x))$, where $f_X(\cdot)$ is the marginal density of the covariate X . Hence this type of variable bandwidth gives the opportunity to change the degree of dependence on the design density $f_X(\cdot)$. For example, $\hat{h}_k^{1/4}(x)$ corresponds to a variable bandwidth that is asymptotically proportional to $f_X^{1/4}(x)$, leading to a smoothing spline-type of design adaptation. The parameter k is used to tune the amount of smoothing.

Theorem 5.1. Suppose that $f_X(\cdot)$ is positive and continuous on a compact interval $[a, b]$ and that $k_n \rightarrow \infty$ such that $k_n/n \rightarrow 0$. Then $\hat{h}_{k_n}(x) = [k_n/(nf_X(x))](1 + o_P(1))$ uniformly in $x \in [a, b]$.

We next examine the limiting distribution for the “ideal” transformed data (4), which have conditional variance $\sigma^{*2}(x) = \text{var}(Y^*|X = x)$. Let $\hat{m}(x; \phi_1, \phi_2)$ be the regression estimator (14) based on the transformed data (4). Let $K(\cdot)$ be a compactly supported probability density function with mean 0, and put $c_K = \int_{-\infty}^{+\infty} v^2 K(v) dv$ and $d_K = \int_{-\infty}^{+\infty} K^2(v)$

$\times dv$. Further, assume that $K(\cdot)$ is uniformly Lipschitz continuous. Then we have the following result.

Theorem 5.2. Suppose that $f_X(\cdot)$, $m''(\cdot)$ and $\sigma^*(\cdot)$ are bounded functions, continuous at the point x , and that $f_X(x) > 0$. If $k_n \rightarrow \infty$ such that $k_n/n \rightarrow 0$, then, conditionally on the covariates $\{X_1, \dots, X_n\}$,

$$\sqrt{k_n}(\hat{m}(x; \phi_1, \phi_2) - m(x) - m''(x)c_K h_k^2(x)/2) \rightarrow N(0, d_K \sigma^{*2}(x)), \quad (15)$$

where $h_k(x) = k_n/(nf_X(x))$, provided that $E(|Y^*|^{2+\gamma} | X = x)$ is continuous at the point x for some $\gamma > 0$.

Finally, we study the asymptotic behavior of the regression estimator based on the estimated transformations. More precisely, we focus our attention on $\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2)$, where $\hat{\phi}_1$ and $\hat{\phi}_2$ estimate ϕ_1 and ϕ_2 . A basic requirement in the consistency result for $\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2)$ is that the estimates $\hat{\phi}_1(t, z)$ and $\hat{\phi}_2(t, z)$ are consistent uniformly for t in a neighborhood of x and for z in an interval chosen such that the instability of $\hat{G}_n$ in the tail can be dealt with. Formally, assume that

$$\beta_n(x) = \max_{j=1,2} \left\{ \sup_{z \in (0, \tau_n), t \in (x \pm \tau)} |\hat{\phi}_j(t, z) - \phi_j(t, z)| \right\} = o_P(1), \quad (16)$$

with $\tau_n > 0$ and $\tau > 0$. Then extend the definition of $\hat{\phi}_j$ ($j = 1, 2$) as follows:

$$\begin{aligned} \hat{\phi}_j(t, z) &= \hat{\phi}_j(t, z), & \text{if } z \leq \tau_n \\ &= z & \text{elsewhere.} \end{aligned} \quad (17)$$

In other words, as already proposed in Section 3, do not transform the data in the region of instability. This approach is justified as long as the contribution from the tail is negligible in the following sense:

$$\begin{aligned} \kappa_n(x) &= \max_{j=1,2} \left\{ \sup_{t \in (x \pm \tau)} E \left(1_{[Z > \tau_n]} |Z - \phi_j(t, Z)| \middle| X = t \right) \right\} \\ &= o(1). \end{aligned}$$

Theorem 5.3. Assume that the conditions of Theorem 5.1 hold. Then

$$\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2) - \hat{m}(x; \phi_1, \phi_2) = O_P(\beta_n(x) + \kappa_n(x)), \quad (18)$$

provided that K is uniformly Lipschitz continuous and has a compact support.

As a consequence of Theorems 5.2 and 5.3, we obtain that $\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2)$ is a consistent estimator of $m(x)$ for any consistent estimators $\hat{\phi}_1$ and $\hat{\phi}_2$. Let $\tau_F(x) = \sup_y \{y : \bar{F}(y|x) < 1\}$ and define $\tau_G(x)$ similarly. If $\tau_F(x') < \tau_G(x')$ for x' in a neighborhood of x , then $\kappa_n(x) = 0$ by taking $\tau_n = \tau(x')$ with $\tau_F(x') < \tau(x') < \tau_G(x')$. With the optimal choice of $k \sim n^{4/5}$, the local average estimator (7) is of uniform rate $(\log n/n)^{2/5}$; that is, $\beta_n(x) = O_P((\log n/n)^{2/5})$. Thus the resulting estimator $\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2)$ has the same rate, using Theorems 5.2 and 5.3.

Next, consider the second type of transformation (10). If, further, the censoring time C is independent of the covariate X , then the estimators $\hat{\phi}_1$ and $\hat{\phi}_2$ are obtained by substituting the Kaplan-Meier estimator into (10). In this case, $\beta_n(x) = O_P((\log n/n)^{1/2})$. Hence the difference in (18) is negligible, implying that the estimator is asymptotically normal,

$$\sqrt{k_n}(\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2) - m(x) - m''(x)c_K h_k^2(x)/2) \rightarrow N(0, d_K \sigma^{*2}(x)),$$

provided that $k_n \log n/n \rightarrow 0$. Thus the rate of convergence is $O_P(n^{-2/5})$ when k_n is chosen optimally.

6. EXAMPLES

In this section we illustrate the proposed estimation method via three examples. In a first example we use a simulated data set to evaluate the performance of the transformations (7) and (10). We then apply the method to the Stanford Heart Transplant Data, given by Miller and Halpern (1982). A study of the Primary Biliary Cirrhosis (PBC) data, described by Fleming and Harrington (1991), is also included to illustrate how the transformation (10) can easily be applied to the multivariate setting. In all the presented examples, we take K to be the standard normal density function.

Example 1: Simulation study. We simulated 200 data points from the following model:

$$\begin{aligned} Y_i &= 4.5 - 64X_i^2(1 - X_i)^2 - 16(X_i - 0.5)^2 + 0.25\varepsilon_i, \\ X_i &\sim \text{iid Uniform}[0, 1], \quad \varepsilon_i \sim \text{iid } N(0, 1), \end{aligned}$$

where $\{X_i\}$ and $\{\varepsilon_i\}$ are independent. The censoring time C_i is conditional independent of the survival time Y_i given X_i and is distributed as $(C_i | X_i = x) \sim_{\text{independent}} \text{exponential}(c(x))$, where $c(x)$ is the mean conditional censoring time given by

$$\begin{aligned} c(x) &= 3(1.25 - |4x - 1|), & \text{if } 0 \leq x \leq .5, \\ &= 3(1.25 - |4x - 3|), & \text{if } .5 < x \leq 1. \end{aligned}$$

In this example, approximately 40% of the 200 observations are censored. The cross-validation curve, shown in Figure 1c, has a flat part in the range of 6–8, which led us to choose $\hat{k} = 7$ as an optimal smoothing parameter. The transformed data, obtained via (1), together with the local linear smoother, calculated via (2), can be found in Figure 1d. For simulated data, we can actually compare this resulting smoother with the smoother based on the unobserved data, which is also presented in Figure 1d. The two smoothers (dashed and dotted curves) are almost indistinguishable, reflecting the effectiveness of the procedure. We also experimented and directly smoothed the censored data in Figure 1b, and found that the resulting smoother clearly underestimates the unknown regression curve. This emphasizes the necessity of transforming the data. The aim of Figure 1, e and f, is to compare the performance of the proposed procedure based on transformation (10). We used $k = 7$ to calculate (12) and the local linear regression smoother (14); we also tried other reasonable values of k , but did not find sig-

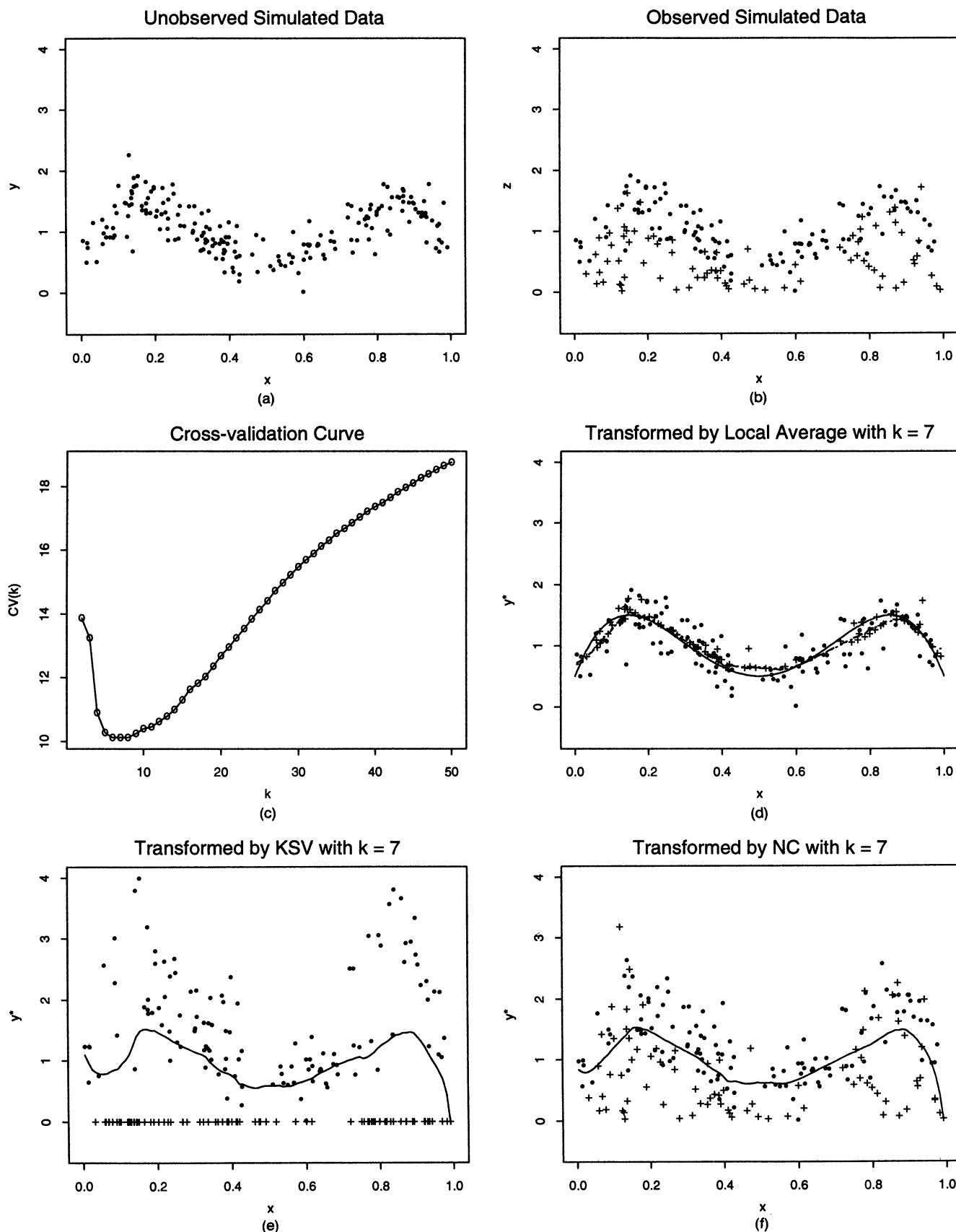


Figure 1. Simulated Data Set. The character "+" indicates the censored observations, and the uncensored observations are presented by ".". The same characters are used for the estimated transformed data. The individual panels show (a) unobserved simulated data; (b) observed simulated data; (c) cross-validation curve; (d) data transformed by local average using an optimal $k=7$ and local linear smoothers: solid curve—the true regression function, dashed curve—the smoother based on the transformed data, dotted curve—the smoother based on the unobserved data in (a); (e) KSV-type of transformed data, together with the local linear smoother, based on $k=7$; and (f) data transformed by the NC-type of transformation, together with the local linear smoother for the transformed data, with $k=7$.

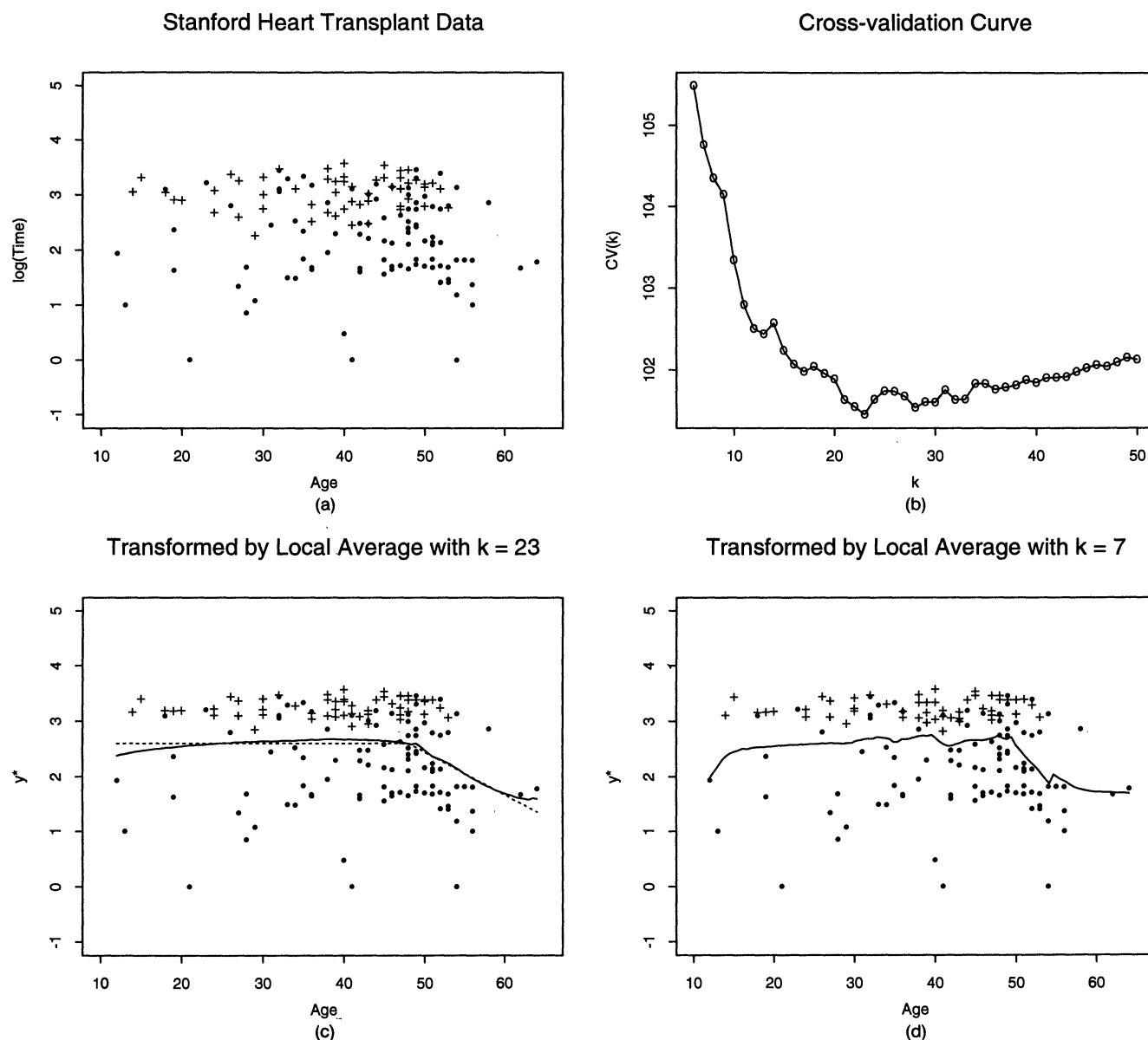


Figure 2. Stanford Heart Transplant Data. The same characters are used as in Figure 1. The individual panels show (a) Stanford Heart Transplant data with log-survival time plotted against age; (b) cross-validation curve; (c) transformation by local average and the resulting estimated regression curve, based on the optimal $\hat{k} = 23$: solid curve—the nonparametric estimated curve, dotted curve—the suggested relationship $2.74 - .078(x - 48)_+$ [see (18)]; and (d) data transformed by local average and the resulting estimated regression curve, based on $k = 7$.

nificantly different resulting transformations and estimated curves.

Note that the scatterplots of the transformed data obtained via transformation (10) show a larger variability than do those obtained via the local average transformation (7). Clearly, the KSV transformation (see Fig. 1e) has a large variance. Moreover, the uncensored observations are transformed into too-large values. (We remark that several large values of the transformed response fall out of the range of the scatterplot). Transformation (10) with $\hat{\alpha} = .15$ [see (11)], presented in Figure 1f, is clearly an improvement over the KSV transformation, both theoretically and empirically.

Example 2: Stanford Heart Transplant data. We now illustrate our method using data from the Stanford Heart Transplant program. Between October 1967 and February

1980, 184 of the 249 patients admitted to this program received a heart transplantation. We focus on only two variables, with the survival time as the response and age as the covariate. Patients alive beyond February 1980 were considered censored. For purposes of comparison with the work of Miller and Halpern (1982) and Doksum and Yandell (1982), we concentrate our analysis on the 157 patients out of 184 who had complete tissue typing. Among the 157 cases, 55 were censored.

We first of all took the optimal $\hat{k} = 23$, obtained from Figure 2b, to transform the data via the local average transformation. The resulting transformed data and its local linear smoother can be found in Figure 2c. Note the flat behavior of the estimated curve for age 10–48. From age 48 on, the curve decreases linearly with a slope of approximately $-.078$

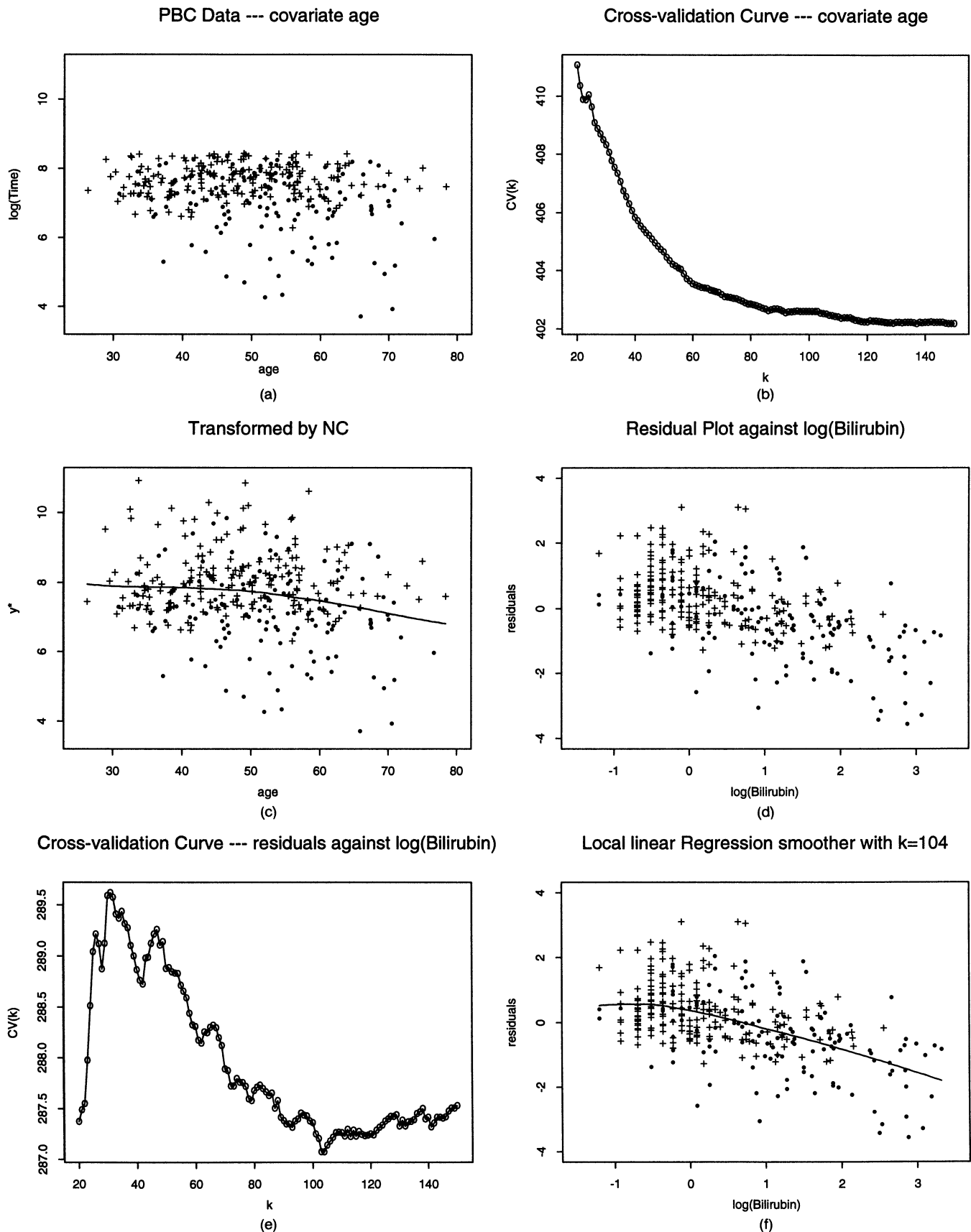


Figure 3. Primary Biliary Cirrhosis Data. The same characters as before are used. The individual panels show (a) scatterplot of $\log(\text{time})$ against age; (b) cross-validation curve with covariate age; (c) transformation by (10) and the resulting local linear smoother based on $\hat{k} = 93$; (d) scatterplot of residuals in Figure 3c against $\log(\text{bilirubin})$; (e) cross-validation curve for the data points in Figure 3d; and (f) local linear regression smoother based on the optimal $\hat{k} = 104$.

and with intercept 2.74 at age 48. This reflects the fact that for earlier age, the log-survival time is nearly independent of age, but at later age it decreases linearly with aging. In terms of hazard rates, this means that the transplantation risk is approximately the same for all patients under age 48 and increases from age 48 on in the following way:

$$h(t|x) = h_0(t\psi(x))\psi(x)$$

with

$$\psi(x) = \exp(-2.74 + .078(x - 48)_+), \quad (18)$$

where h_0 is the so-called baseline hazard function. Such a relation appears to be new. The result supports our intuition and, moreover, gives a deeper insight into the heart transplantation risk at various ages. In comparison with previous studies by, for example, Miller and Halpern (1982) and Doksum and Yandell (1982), our analysis gives a more precise description of the data structure. In an attempt to search for finer structure, we estimated the unknown regression curve using a smaller smoothing parameter $k = 7$. The result is presented in Figure 2d.

Example 3: Primary Biliary Cirrhosis (PBC) data. We use this example to illustrate how the transformation (10) can easily be used in the multivariate setting. Between January 1974 and May 1984, the Mayo Clinic collected data on PBC, a rare but fatal chronic liver disease of unknown cause. Of the 424 registered patients, 312 participated in the randomized trial; our analysis is based on those patients. (For a detailed description and an analysis of the data set, see Fleming and Harrington 1991, p. 2, chap. 4). To illustrate our methodology, we concentrate on only three variables. As the response variable, we take the logarithm of the time (in days) between registration and death, or liver transplantation or time of the study analysis (July 1986). Of the 312 patients, 187 cases were censored. We study the effect of the covariates age and logarithm of bilirubin.

We first use the NC transformation given in (10) to account for the censoring. Then we apply the local linear regression smoother (14) with a cross-validated smoothing parameter to study the relationship between the survival time and age. The results are presented in Figure 3, a-c. After adjusting the age effect, we exploit the effect of log(bilirubin) on log(time), based on the residuals from the first part of the analysis. Figure 3, d-f, reports this analysis.

APPENDIX: PROOFS

Proof of Theorem 5.1

Let $F_X(\cdot)$ denote the marginal distribution function of X . In this proof we suppress the dependence on n in k_n . We will show that

$$\max_{k < a \leq n-k} \left| \frac{F_X(X_{a+k}) - F_X(X_{a-k})}{2k/n} - 1 \right| = o_P(1). \quad (A.1)$$

From this statement, we proceed as follows. By the mean value theorem,

$$F_X(X_{a+k}) - F_X(X_{a-k}) = 2\hat{h}_k(x)f_X(\xi_n(x)), \quad (A.2)$$

where $\xi_n(x) \in [X_{a-k}, X_{a+k}]$. Now from (A.1) we obtain that $\sup_{x \in [a, b]} |\xi_n(x) - x| = o_P(1)$. Hence we can conclude from (A.2) that

$$\hat{h}_k(x) = \frac{F_X(X_{a+k}) - F_X(X_{a-k})}{2f_X(x)} (1 + o_P(1))$$

uniformly in $x \in [a, b]$. Thus the result follows from (A.1).

We now prove (A.1). From the representation of uniform order statistics, we have $F_X(X_{a+k}) - F_X(X_{a-k}) = \sum_{j=a-k+1}^{a+k} E_j / \sum_{j=1}^{n+1} E_j$, where $\{E_j: 1 \leq j \leq n+1\}$ is a sequence of iid standard exponentially distributed random variables. Hence (A.1) is equivalent to

$$\max_{k < a \leq n-k} \left| \sum_{j=a-k+1}^{a+k} E_j / (2k) - 1 \right| = o_P(1), \quad (A.3)$$

because $\sum_{j=1}^{n+1} E_j / n = 1 + o_P(1)$ by the weak law of large numbers. To prove (A.3), we must evaluate

$$P \left\{ \left| \sum_{j=a-k+1}^{a+k} E_j / (2k) - 1 \right| \geq \varepsilon \right\} = P \left\{ \sum_{j=a-k+1}^{a+k} E_j \leq (1 - \varepsilon)2k \right\} + P \left\{ \sum_{j=a-k+1}^{a+k} E_j \geq (1 + \varepsilon)2k \right\}$$

for any $\varepsilon > 0$. Because $\sum_{j=a-k+1}^{a+k} E_j$ is gamma($2k$) distributed, it can be shown that

$$P \left\{ \sum_{j=a-k+1}^{a+k} E_j \leq (1 - \varepsilon)2k \right\} = \sum_{i=2k}^{\infty} \frac{(2k(1 - \varepsilon))^i}{i!} \exp(-2k(1 - \varepsilon)). \quad (A.4)$$

Note that for $i \geq 2k$, the sequence $(2k(1 - \varepsilon))^i / (i - 2)!$ is decreasing. Therefore, (A.4) can be bounded by

$$\begin{aligned} & \frac{(2k(1 - \varepsilon))^{2k}}{(2k - 2)!} \sum_{i=2k}^{\infty} \frac{1}{i(i - 1)} \exp(-2k(1 - \varepsilon)) \\ & \leq \frac{1}{\sqrt{2\pi e}} \sqrt{2k - 1} (1 - 1/2k)^{-2k} \exp(2k(\varepsilon + \log(1 - \varepsilon))) \\ & = o(\exp(-k\varepsilon^2/2)), \end{aligned} \quad (A.5)$$

where we used Stirling's formula and the fact that $\log(1 - \varepsilon) < -\varepsilon - \varepsilon^2/2$. Using similar arguments, we can show that $P\{\sum_{j=a-k+1}^{a+k} E_j \geq (1 + \varepsilon)2k\} = o(\exp(-k\varepsilon^2/2))$. This, together with (A.5), leads to $P\{|\sum_{j=a-k+1}^{a+k} E_j / (2k) - 1| \geq \varepsilon\} = o(\exp(-k\varepsilon^2/2))$. Hence (A.3) follows from

$$P \left\{ \max_{k < a \leq n-k} \left| \sum_{j=a-k+1}^{a+k} E_j / (2k) - 1 \right| \geq \varepsilon \right\} = o(2k \exp(-k\varepsilon^2/2)),$$

as had to be shown.

Proof of Theorem 5.2

Also in this proof, we suppress the dependence on n in k_n . Let $h_k = k/(nf_X(x))$, and let $\hat{m}_0(x)$ be defined by (14), but now with $\hat{h}_k(x)$ replaced by h_k . Applying theorem 5 of Fan and Gijbels (1992) together with the consistent result in our Theorem 5.1, we have $E((\hat{m}(x; \phi_1, \phi_2) - \hat{m}_0(x))^2 | \mathbf{X}) = o_P((k/n)^4 + 1/k)$, where $\mathbf{X} = \{X_1, \dots, X_n\}$. Therefore, we need only establish the conditional asymptotic normality of $\hat{m}_0(x)$. Recall that $m(x) = E(Y^* | X = x)$. To show the conditional asymptotic normality,

$$\begin{aligned} & \frac{\hat{m}_0(x) - E\{\hat{m}_0(x) | \mathbf{X}\}}{\sqrt{\text{var}(\hat{m}_0(x) | \mathbf{X})}} \\ & = \frac{\sum_{i=1}^n w_i(x)(Y_i^* - m(X_i))}{\sum_{i=1}^n w_i(x) \sqrt{\text{var}(\hat{m}_0(x) | \mathbf{X})}} \rightarrow N(0, 1), \end{aligned} \quad (A.6)$$

it is sufficient to check Lyapounov's condition,

$$\frac{1}{\sigma_n^{2+\gamma}} \sum_{i=1}^n E(|w_i(x)(Y_i^* - m(X_i))|^{2+\gamma} | \mathbf{X}) \rightarrow 0, \quad (\text{A.7})$$

where $\sigma_n = \sum_{i=1}^n w_i(x) \sqrt{\text{var}(\hat{m}_0(x) | \mathbf{X})}$. Because K has a bounded support, we have

$$\begin{aligned} E \left[\sum_{i=1}^n E(|w_i(x)(Y_i^* - m(X_i))|^{2+\gamma} | \mathbf{X}) \right] \\ = n E |w_1(x)(Y_1^* - m(X_1))|^{2+\gamma} \\ = n O \left(E \left[\left| K \left(\frac{x - X_1}{h_k} \right) (s_{n,2} - s_{n,1}(x - X_1)) \right|^{2+\gamma} \right] \right). \quad (\text{A.8}) \end{aligned}$$

Straightforward arguments yield

$$\begin{aligned} E \left\{ \left| K \left(\frac{x - X_1}{h_k} \right) s_{n,2} \right|^{2+\gamma} \right\} \\ \leq 2^{1+\gamma} E \left[K^2 \left(\frac{x - X_1}{h_k} \right) (x - X_1)^2 \right]^{2+\gamma} \\ + 2^{1+\gamma} E K^{2+\gamma} \left(\frac{x - X_1}{h_k} \right) E \left[\sum_{i=2}^n K \left(\frac{x - X_i}{h_k} \right) (x - X_i)^2 \right]^{2+\gamma} \\ = O(n^{2+\gamma} h_k^{3(2+\gamma)+1}). \quad (\text{A.9}) \end{aligned}$$

The other term in (A.8) can be dealt with similarly and has the same order as (A.9). Therefore, (A.8) is of the order $O(n_k^{3+\gamma} h_k^{3(2+\gamma)+1})$. This implies that

$$\sum_{i=1}^n E(|w_i(x)(Y_i^* - m(X_i))|^{2+\gamma} | \mathbf{X}) = O_P(n^{3+\gamma} h_k^{3(2+\gamma)+1}). \quad (\text{A.10})$$

As was shown by Fan and Gijbels (1992),

$$\sum_{i=1}^n w_i(x) = s_{n,0} s_{n,2} - s_{n,1}^2 = c_K f_X^2(x) n^2 h_k^4 (1 + o_P(1)),$$

$$E(\hat{m}_0(x) | \mathbf{X}) = m(x) + \frac{1}{2} m''(x) c_K h_k^2 + o_P(h_k^2),$$

$$\text{var}(\hat{m}_0(x) | \mathbf{X}) = d_K \sigma^{*2}(x) / k + o_P(1/k). \quad (\text{A.11})$$

Hence $\sigma_n = c n^2 h_k^4 k^{-1/2} (1 + o_P(1))$, where c is a positive constant. This, together with (A.10), justifies the Lyapounov condition (A.7). Finally, the result (15) follows from (A.6) by substituting (A.11).

Proof of Theorem 5.3

We first remark that in the sequel, $\hat{h}_k(x)$ can always be replaced by $k/(nf_X(x))$, if necessary, by using arguments analogous to those in the proof of Theorem 5.2. Recall the unestimated transformed data (4) and denote the estimated transformed data by $\hat{Y}^* = \delta \hat{\phi}_1(X, Z) + (1 - \delta) \hat{\phi}_2(X, Z)$, where $\hat{\phi}_1$ and $\hat{\phi}_2$ are given in (17). It follows from (14) that

$$\begin{aligned} |\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2) - \hat{m}(x; \phi_1, \phi_2)| \\ \leq \sum_{i=1}^n |w_i(x)| \|\hat{Y}_i^* - Y_i^*\| \left/ \sum_{i=1}^n w_i(x) \right|. \quad (\text{A.12}) \end{aligned}$$

Note that

$$\frac{\sum_{i=1}^n |w_i(x)|}{\sum_{i=1}^n w_i(x)} \leq \frac{2s_{n,2}s_{n,0}}{s_{n,2}s_{n,0} - s_{n,1}^2} = 2 + o_P(1). \quad (\text{A.13})$$

Because K has a bounded support, the effective design points are actually those local to the point x . Using this fact, and combining (A.12) and (A.13), we find that

$$\begin{aligned} |\hat{m}(x; \hat{\phi}_1, \hat{\phi}_2) - \hat{m}(x; \phi_1, \phi_2)| \leq 2\beta_n(x)(1 + o_P(1)) \\ + \frac{1}{\sum_{i=1}^n w_i(x)} \sum_{i=1}^n \sum_{j=1}^2 |w_i(x)| 1_{[Z_i > \tau_n]} |Z_i - \phi_j(X_i, Z_i)|, \quad (\text{A.14}) \end{aligned}$$

where $\beta_n(x)$ is given by (16). Now

$$\begin{aligned} E \left[\frac{1}{\sum_{i=1}^n w_i(x)} \sum_{i=1}^n \sum_{j=1}^2 |w_i(x)| 1_{[Z_i > \tau_n]} |Z_i - \phi_j(X_i, Z_i)| \middle| X_1, \dots, X_n \right] \\ \leq 2\kappa_n(x) \frac{\sum_{i=1}^n |w_i(x)|}{\sum_{i=1}^n w_i(x)}, \end{aligned}$$

and hence the result follows from (A.13) and (A.14).

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REFERENCES

- Beran, R. (1981), "Nonparametric Regression With Randomly Censored Survival Data," technical report, Department of Statistics, University of California, Berkeley.
- Buckley, J., and James, I. R. (1979), "Linear Regression with Censored Data," *Biometrika*, 66, 429–436.
- Cleveland, W. S. (1979), "Robust Locally Weighted Regression and Smoothing Scatterplots," *Journal of the American Statistical Association*, 74, 829–836.
- Dabrowska, D. M. (1987), "Nonparametric Regression With Censored Data," *Scandinavian Journal of Statistics*, 14, 181–197.
- Doksum, K. A., and Yandell, B. S. (1982), "Properties of Regression Estimates Based on Censored Survival Data," in *A Festschrift for Erich Lehmann*, eds. P. J. Bickel, K. A. Doksum, and J. L. Hodges, Jr., Belmont, CA: Wadsworth International Group, pp. 140–156.
- Eubank, R. L. (1988), *Spline Smoothing and Nonparametric Regression*, New York: Marcel Dekker.
- Fan, J. (1992), "Design-Adaptive Nonparametric Regression," *Journal of the American Statistical Association*, 87, 998–1004.
- (1993), "Local Linear Regression Smoothers and Their Minimax Efficiency," *The Annals of Statistics*, 21, 196–216.
- Fan, J., and Gijbels, I. (1992), "Variable Bandwidth and Local Linear Regression Smoothers," *The Annals of Statistics*, 20, 2008–2036.
- Fan, J., and Marron, J. S. (1994), "Fast Implementations of Nonparametric Curve Estimators," submitted to *Journal of Computational and Graphical Statistics*.
- Fleming, T. R., and Harrington, D. P. (1991), *Counting Processes and Survival Analysis*, New York: John Wiley.
- Gasser, T., and Müller, H. G. (1979), "Kernel Estimation of Regression Functions," in *Smoothing Techniques for Curve Estimation* (Lecture Notes in Mathematics 757), New York: Springer-Verlag, pp. 23–68.
- Härdle, W. (1990), *Applied Nonparametric Regression*, Cambridge, U.K.: Cambridge University Press.
- James, I. R., and Smith, P. J. (1984), "Consistency Results for Linear Regression With Censored Data," *The Annals of Statistics*, 12, 590–600.
- Koul, H., Susarla, V., and Van Ryzin, J. (1981), "Regression Analysis With Randomly Right-Censored Data," *The Annals of Statistics*, 9, 1276–1288.
- Lai, T. L., and Ying, Z. (1991), "Large-Sample Theory of a Modified Buckley-James Estimator for Regression Analysis With Censored Data," *The Annals of Statistics*, 19, 1370–1402.
- Leurgans, S. (1987), "Linear Models, Random Censoring, and Synthetic Data," *Biometrika*, 74, 301–309.
- Miller, R., and Halpern, J. (1982), "Regression With Censored Data," *Biometrika*, 69, 521–531.
- Müller, H.-G. (1988), *Nonparametric Analysis of Longitudinal Data*, Berlin: Springer-Verlag.
- Nadaraya, E. A. (1964), "On Estimating Regression," *Theory of Probability and Its Applications*, 9, 141–142.
- Ritov, Y. (1990), "Estimation in a Linear Regression Model With Censored Data," *The Annals of Statistics*, 18, 303–328.
- Silverman, B. W. (1986), *Density Estimation for Statistics and Data Analysis*, London: Chapman and Hall.
- Staniswalis, J. G. (1989), "Local Bandwidth Selection for Kernel Estimates," *Journal of the American Statistical Association*, 84, 284–288.
- Stone, C. J. (1977), "Consistent Nonparametric Regression," *The Annals of Statistics*, 5, 595–645.

- Tsiatis, A. A. (1990), "Estimating Regression Parameters Using Linear Rank Tests for Censored Data," *The Annals of Statistics*, 18, 354–372.
- Wahba, G. (1990), *Spline Models for Observational Data*, Philadelphia: SIAM.
- Watson, G. S. (1964), "Smooth Regression Analysis," *Sankhyā*, Ser. A, 26, 359–372.
- Zheng, Z. (1987), "A Class of Estimators of the Parameters in Linear Regression With Censored Data," *Acta Mathematicae Applicatae Sinica*, 3, 231–241.
- (1988), "Strong Consistency of Nonparametric Regression Estimates With Censored Data," *Journal of Mathematical Research and Exposition*, 8, 307–313.